NoiseBridge: Phonetic-Weighted Contrastive Learning for ASR-Robust Hate Speech Detection in Hindi-English Code-Mixed Text

Abhijit Kumar
Department of Computer Science and Engineering
Indian Institute of Information Technology, Senapati
Manipur, India
abhi230101031@iiitmanipur.ac.in

Dr. N. Kishorjit Singh
Department of Computer Science and Engineering
Indian Institute of Information Technology, Senapati
Manipur, India
kishorjit@iiitmanipur.ac.in

***Abstract*—Hate speech detection systems trained on clean text degrade when applied to noisy transcripts generated by Automatic Speech Recognition (ASR), particularly for code-mixed languages such as Hindi-English (Hinglish) where error rates are substantially higher. We introduce HindiMix-Noisy, the first benchmark for evaluating hate speech detection under controlled multi-level ASR noise in code-mixed text, comprising 123,631 training examples across four noise conditions.**

We propose NoiseBridge, a noise-robust training framework built around PWNIC (Phonetic-Weighted Noise-Invariant Contrastive) loss, which scales contrastive alignment according to phonetic dissimilarity measured via Jaro-Winkler distance [1]. NoiseBridge further incorporates a noise-aware attention gate and an auxiliary noise-level prediction objective.

Across three multilingual encoders (mBERT, MuRIL, XLM-R), NoiseBridge achieves consistent improvements on XLM-R (+0.41 macro-F1 on average, +0.56 under high noise; averaged over 3 seeds) while preserving baseline performance on others. Our analysis reveals that multilingual transformers already exhibit partial robustness to character-level noise, but targeted contrastive alignment provides additional gains under severe corruption. We further demonstrate that adversarial noise disentanglement via gradient reversal [2] fails catastrophically in this setting, motivating our simplified design.

These findings highlight the importance of phonetic-aware objectives for robust NLP in code-mixed and ASR-degraded scenarios.

***Index Terms*—hate speech detection, code-mixed text, ASR noise robustness, contrastive learning, phonetic similarity, multilingual NLP**

I. Introduction

Online hate speech detection has made substantial progress with transformer-based models fine-tuned on labeled datasets [3], [4]. However, the majority of existing benchmarks assume clean, well-formed text input. In practice, a growing fraction of online content originates from speech—voice messages, live-streamed audio, and podcast transcripts—which is processed by Automatic Speech Recognition (ASR) systems before being analyzed.

ASR output is inherently noisy: character substitutions, insertions, deletions, and word boundary errors are common, particularly for *code-mixed* languages where speakers alternate between two or more languages within an utterance. Hindi-English code-mixing ("Hinglish") is the most prevalent form of code-mixing on the Indian internet [5], with an estimated 350 million speakers. Current ASR systems handle Hinglish poorly, producing transcripts with error rates of 15–40% [6], compared to 5–10% for monolingual English.

This raises a fundamental question: *how robust are existing hate speech detectors to ASR-induced noise on code-mixed text, and can we improve that robustness?* We address this with three contributions:

1) **HindiMix-Noisy Benchmark**: the first multi-level ASR noise benchmark for hate speech detection on Hindi-English code-mixed text, with 123,631 training examples and four test conditions (clean, low, medium, high noise).
2) **NoiseBridge**: a noise-robust training framework centered on PWNIC (Phonetic-Weighted Noise-Invariant Contrastive) loss, which weights contrastive alignment pressure by Jaro-Winkler phonetic dissimilarity between clean and noisy versions of each utterance.
3) **Empirical Analysis**: systematic experiments with three multilingual encoders (mBERT, MuRIL, XLM-R), three seeds, and four noise conditions, including an ablation documenting the catastrophic failure of adversarial gradient reversal in this setting.

II. Related Work

Hate Speech Detection. Transformer-based approaches dominate recent benchmarks [7], [8]. Multilingual models like mBERT [9], XLM-R [10], and MuRIL [11] enable cross-lingual transfer, but evaluation has focused on clean text.

Code-Mixed NLP. Hindi-English code-mixed text poses challenges for tokenization [12] and semantic understanding [13]. Prior work on code-mixed hate speech [14], [15] has not studied the impact of ASR-induced noise.

Noise-Robust NLP. Character-level noise robustness has been studied for NER [16], machine translation [17], and sentiment analysis [18]. ByT5 [19] provides byte-level encoding that avoids subword OOV issues. No prior work addresses

noise robustness specifically for hate speech on code-mixed text.

Contrastive and Adversarial Learning. SimCSE [20] and supervised contrastive learning [21] have been used for representation learning. We extend InfoNCE [22] with phonetic weighting. The closest related work to ours is SpokenCSE [23], which applies contrastive pre-training between clean and ASR-noisy transcripts to improve robustness in spoken language understanding. However, SpokenCSE applies *uniform* contrastive pressure regardless of how distorted a pair is, and targets intent detection in English, not hate speech in code-mixed text. PWNIC directly addresses this gap by weighting each pair's contrastive contribution by its phonetic dissimilarity. Domain adversarial neural networks (DANN) [2] use gradient reversal for domain-invariant representations; we explore this for noise disentanglement and document its failure (Section VII-B).

III. HindiMix-Noisy Benchmark

A. Data Collection

We aggregate six publicly available hate speech datasets into a unified corpus: Davidson et al. [3] (17,290 tweets), OLID [24] (9,774), SemEval 2020 Task 9 (18,631 Hindi-English tweets), HASOC 2021 [15] (7,952 Hindi-English tweets), UCB Measuring Hate (27,854), TweetEval Hate (8,909) and web scraped (34,293). All labels are unified into a binary schema (`hate`/`non-hate`). The final dataset consists of 123,631 training examples, comprising a clean subset of 78,778 instances (37.6% hate) and three synthetically generated noisy subsets (low, medium, high), each containing 14,951 examples with 6.3% hate. The corpus includes both English and Hindi-English (code-mixed) content. Table I summarizes the benchmark statistics.

TABLE I
HindiMix-Noisy Benchmark Statistics

Split	Examples	Hate %	Languages
Train (clean)	78,778	37.6	en, hi-en
Train (low)	14,951	6.3	en, hi-en
Train (medium)	14,951	6.3	en, hi-en
Train (high)	14,951	6.3	en, hi-en
Validation	16,882	37.6	en, hi-en
Test (each condition)	16,882	37.6	en, hi-en

B. Noise Injection

We simulate ASR-induced noise by applying character-level perturbations calibrated to empirical ASR error rates on Hinglish audio [6]. Let $x = (c_1, c_2, \ldots, c_n)$ be a clean text. For noise level $\ell \in \{\texttt{low}, \texttt{medium}, \texttt{high}\}$ with perturbation probability $p_\ell \in \{0.05, 0.10, 0.20\}$, we apply to each character c_i one of four operations with probability $p_\ell/4$ each:

$$c_i \to c_i' \sim \mathcal{U}(\Sigma) \qquad \text{(substitution)} \tag{1}$$

$$c_i \to c_i \cdot c' \sim \mathcal{U}(\Sigma) \qquad \text{(insertion)} \tag{2}$$

$$c_i \to \epsilon \qquad \text{(deletion)} \tag{3}$$

$$c_i \to c_i \cdot \texttt{` '} \qquad \text{(word split)} \tag{4}$$

where Σ is the character vocabulary and ϵ is the empty string.

IV. Proposed Method: NoiseBridge

NoiseBridge augments a pretrained multilingual encoder with three components: (1) a noise-aware attention gate, (2) PWNIC contrastive loss, and (3) an auxiliary noise-level predictor.

A. Noise-Aware Attention Gate

Given contextual representations $\mathbf{H} = f_\theta(x) \in \mathbb{R}^{L \times d}$ from a pretrained encoder f_θ, we apply a learned sigmoid gate to each token:

$$g_\psi(\mathbf{H}) = \mathbf{H} \odot \sigma(\mathbf{W}_g \mathbf{H} + \mathbf{b}_g) \tag{5}$$

where $\mathbf{W}_g \in \mathbb{R}^{d \times 1}$, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication. This gate implicitly learns to down-weight tokens corrupted by ASR noise. The gated representations are mean-pooled with attention masking:

$$\mathbf{z} = \frac{\sum_{i=1}^{L} g_\psi(\mathbf{h}_i) \cdot m_i}{\sum_{i=1}^{L} m_i} \tag{6}$$

where $m_i \in \{0, 1\}$ is the attention mask and $\mathbf{z} \in \mathbb{R}^d$ is the sentence representation.

B. Phonetic-Weighted Noise-Invariant Contrastive Loss

PWNIC aligns clean and noisy representations of the same utterance in a learned projection space, with alignment pressure proportional to phonetic dissimilarity.

Phonetic Dissimilarity. For a clean–noisy pair $(x^c, \tilde{x})$, we compute word-level phonetic dissimilarity:

$$\phi(x^c, \tilde{x}) = 1 - \frac{1}{M} \sum_{j=1}^{M} \mathrm{JW}(w_j^c, w_j^n) \tag{7}$$

where $M = \max(|x^c|_w, |\tilde{x}|_w)$ is the maximum word count, $\mathrm{JW}(\cdot, \cdot)$ is the Jaro-Winkler similarity [1], and unpaired words contribute similarity 0. Thus $\phi \in [0, 1]$: high ϕ indicates severe phonetic distortion requiring stronger contrastive alignment.

PWNIC Objective. Let $\mathbf{p}_i^c = \pi_\xi(\mathbf{z}_i^c)$ and $\mathbf{p}_i^n = \pi_\xi(\mathbf{z}_i^n)$ be projected representations via a two-layer MLP π_ξ. For a batch of N clean–noisy pairs:

$$\mathcal{L}_{\text{PWNIC}} = -\frac{1}{\sum_i \phi_i} \sum_{i=1}^{N} \phi_i \cdot \log \frac{e^{\mathrm{sim}(\mathbf{p}_i^c, \mathbf{p}_i^n)/\tau}}{\sum_{\substack{k=1 \\ k \neq i}}^{2N} e^{\mathrm{sim}(\mathbf{p}_i^c, \mathbf{p}_k)/\tau}} \tag{8}$$

where $\mathrm{sim}(\mathbf{a}, \mathbf{b}) = \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$ is cosine similarity, $\tau = 0.15$ is a temperature parameter, and the denominator sums

over all $2N$ projected embeddings excluding the anchor. This generalizes InfoNCE [22] with instance-dependent weights: when $\phi_i = 0$ (identical texts), the pair contributes zero loss; when ϕ_i is high (severe distortion), the gradient is proportionally stronger. The projection head π_ξ ensures contrastive gradients do not interfere with the classification space [25]. Unlike SpokenCSE [23], which treats all clean–noisy pairs equally regardless of corruption degree, PWNIC allocates stronger alignment pressure to pairs that are more phonetically distant—a property especially important for code-mixed text, where ASR noise interacts unevenly with Hinglish phonetics.

C. Auxiliary Noise-Level Prediction

A separate classifier q_η predicts the noise level from the noisy representation as a multi-task regularizer:

$$\mathcal{L}_{\text{aux}} = -\sum_{k=0}^{K-1} y_k^{\text{noise}} \log q_\eta(\mathbf{z}^n)_k \tag{9}$$

where $K = 4$ noise levels. This head is applied only to $\mathbf{z}^n$ and is **auto-disabled** when fewer than two distinct noise levels are present, preventing degenerate constant-target training (see Section VII-B).

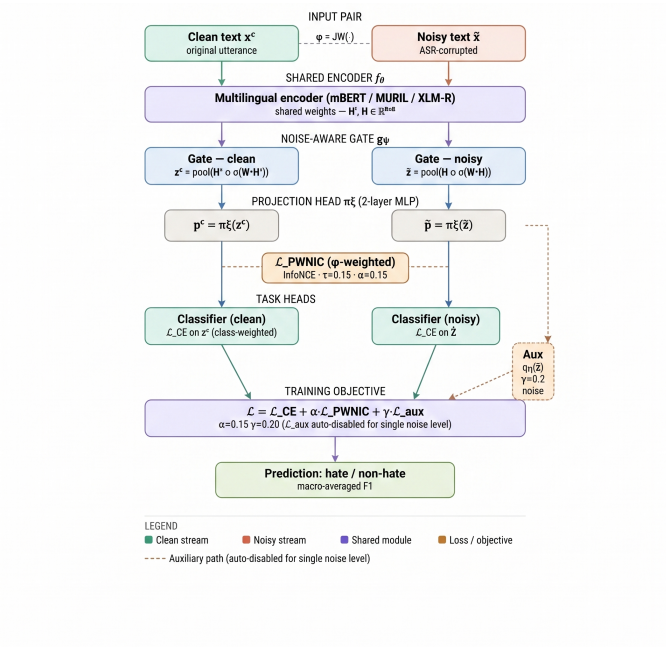

Fig. 1. NoiseBridge architecture

D. Training Objective

The complete loss is:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{CE}}}_{\text{classification}} + \alpha \underbrace{\mathcal{L}_{\text{PWNIC}}}_{\text{contrastive}} + \gamma \underbrace{\mathcal{L}_{\text{aux}}}_{\text{noise pred.}} \tag{10}$$

where $\alpha = 0.15$ and $\gamma = 0.2$. Cross-entropy $\mathcal{L}_{\text{CE}}$ uses class weights inversely proportional to class frequency (1.66:1 imbalance). A per-row CE masking scheme ensures each training example contributes exactly one cross-entropy update per epoch, matching the baseline supervision count exactly.

V. Experimental Setup

Encoders. We evaluate NoiseBridge with three multilingual encoders: mBERT [9] (WordPiece, 104 languages), MuRIL [11] (WordPiece, 17 Indian languages), and XLM-R [10] (SentencePiece, 100 languages). Three lexical baselines are also included: TF-IDF + Logistic Regression, TF-IDF + SVM, and Char-CNN [26].

Training. All transformer models: 5 epochs, AdamW [27] (lr $= 3 \times 10^{-5}$, weight decay 0.01), linear warmup (10% of steps), batch size 16, max length 128, FP16. Both baselines and NoiseBridge train on the full dataset (clean + all noisy subsets), producing one model per encoder evaluated on all four test conditions.

Reproducibility. All experiments use 3 seeds (42, 43, 44) with full seeding (Python random, NumPy, PyTorch CPU/CUDA, cudnn.deterministic). Evaluation metric: macro-averaged F1.

VI. Results

Fig. 2 visualizes performance degradation under increasing noise, while Tables II, III, and IV present the quantitative results.

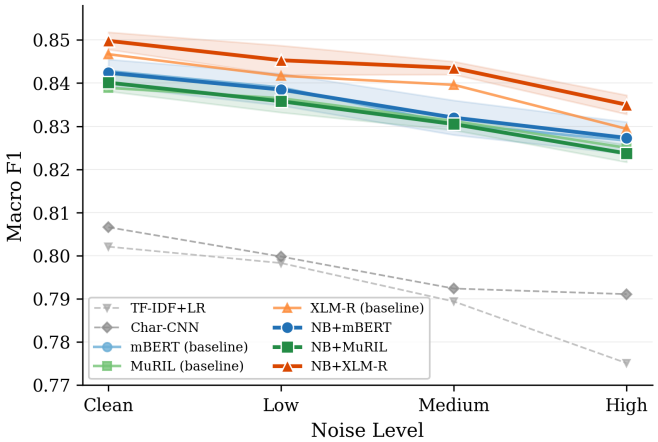

Fig. 2. Performance degradation under increasing ASR noise. XLM-R + NoiseBridge (top curve) exhibits the flattest degradation, with the performance gap over the XLM-R baseline widening at higher noise levels. Shaded bands denote ±1 standard deviation over 3 seeds.

TABLE II
Lexical Baselines (Macro F1)

Model	Clean	Low	Medium	High
TF-IDF + LR	0.8021	0.7983	0.7894	0.7751
TF-IDF + SVM	0.8004	0.7973	0.7896	0.7762
Char-CNN	0.8066	0.7998	0.7924	0.7911

TABLE III
TRANSFORMER BASELINES (MACRO F1)

Model	Clean	Low	Medium	High
mBERT	0.8428	0.8390	0.8305	0.8267
MuRIL	0.8389	0.8365	0.8312	0.8250
XLM-R	0.8467	0.8417	0.8396	0.8294

TABLE IV
NOISEBRIDGE (OURS), MEAN ± STD

Model	Clean	Low	Medium	High
mBERT + NB	0.842±.003	0.839±.004	0.832±.004	0.827±.004
MuRIL + NB	0.840±.002	0.836±.003	0.830±.001	0.824±.002
XLM-R + NB	**0.850**±.002	**0.845**±.003	**0.844**±.002	**0.835**±.002

Finding 1: Transformer encoders are inherently robust. All three transformer baselines degrade by only 1.0–1.7 F1 points from clean to high noise, substantially less than lexical baselines (e.g., TF-IDF+LR drops by 2.70). This suggests that contextual representations effectively absorb character-level perturbations.

Finding 2: NoiseBridge consistently improves XLM-R. XLM-R + NoiseBridge outperforms the XLM-R baseline across all noise conditions, achieving an average gain of $+0.41$ F1. The largest improvement is observed under high noise ($+0.56$), indicating that NoiseBridge is particularly effective in challenging settings. In contrast, gains for mBERT and MuRIL remain within seed variance ($\Delta_{\text{avg}} \approx 0$).

Finding 3: Gains increase with noise severity. For XLM-R, improvements grow monotonically with noise level: $+0.31$ (clean), $+0.36$ (low), $+0.39$ (medium), and $+0.56$ (high). This trend confirms that the proposed PWNIC mechanism provides increasing benefit as noise intensifies.

Why XLM-R benefits more than other models. We hypothesize that XLM-R's SentencePiece tokenizer preserves phonetic structure more effectively than WordPiece tokenization used in mBERT and MuRIL. SentencePiece operates directly on raw Unicode sequences without aggressive normalization, whereas WordPiece applies case folding and accent stripping, potentially discarding phonetic cues critical for handling noisy Hinglish text.

VII. ANALYSIS

A. Component Ablation

Table V and Fig. 3 show the contribution of each NoiseBridge component on XLM-R (seed 42, average F1 across all four test conditions).

TABLE V
ADDITIVE COMPONENT ANALYSIS (XLM-R, AVG F1)

Configuration	α	γ	**Avg F1**
XLM-R baseline (CE only)	–	–	0.8394
+ Noise-aware gate	0	0	0.8339
+ PWNIC	0.15	0	0.8401
+ Aux	0.15	0.2	**0.8434**
Full NB v1 (+ GRL)	0.15	0.1	0.8237

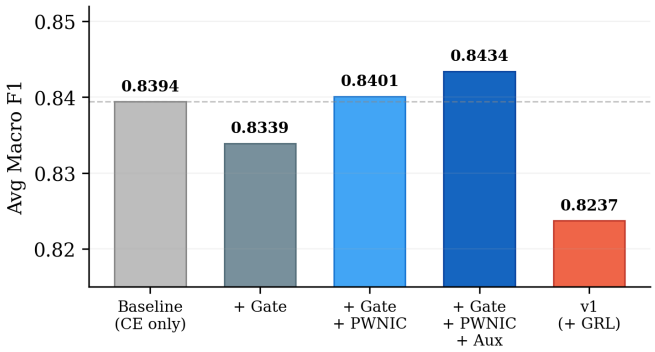

Fig. 3. Component ablation on XLM-R. The noise-aware gate alone slightly hurts (-0.55 from baseline). Adding PWNIC recovers and exceeds baseline ($+0.07$). The auxiliary head provides a further $+0.33$ gain. Adding GRL (v1 design) catastrophically degrades performance (-1.57 from baseline).

An interesting finding is that the noise-aware gate *alone* slightly hurts performance (0.8339 vs. 0.8394 baseline). This suggests that without the contrastive signal from PWNIC to guide it, the gate learns to suppress tokens indiscriminately. PWNIC provides the critical training signal ($+0.62$ over gate-only), and the auxiliary head adds a meaningful further boost ($+0.33$).

B. Adversarial Gradient Reversal Ablation

An earlier version of NoiseBridge (v1) included a gradient reversal layer (GRL) [2] for adversarial noise disentanglement. Table VI and Fig. 4 present this as a negative result.

TABLE VI
V1 (WITH GRL) VS. V2 (WITHOUT GRL) VS. BASELINE

Model	**C**	**L**	**M**	**H**
XLM-R				
Baseline	.847	.842	.840	.829
NB v1 (w/ GRL)	.844	.840	.834	.777
NB v2 (no GRL)	**.850**	**.845**	**.844**	**.835**
MuRIL				
Baseline	.839	.837	.831	.825
NB v1 (w/ GRL)	.832	.834	.786	.815
NB v2 (no GRL)	**.840**	**.836**	**.830**	**.824**
mBERT				
Baseline	.843	.839	.831	.827
NB v1 (w/ GRL)	.832	.830	.822	.818
NB v2 (no GRL)	**.842**	**.839**	**.832**	**.827**

The v1 architecture added a GRL and adversarial noise predictor:

$$\mathcal{L}_{\text{v1}} = \mathcal{L}_{\text{CE}} + \alpha\mathcal{L}_{\text{PWNIC}} + \beta\mathcal{L}_{\text{adv}}^{\text{GRL}} + \gamma\mathcal{L}_{\text{aux}} \tag{11}$$

where the GRL reverses gradients from $\mathcal{L}_{\text{adv}}$ to the encoder.

Failure Mode 1: Degenerate Targets. When training on a single noise level, all noise labels are identical ($y_i^{\text{noise}} = c$ $\forall i$). The adversarial predictor trivially achieves zero loss; the GRL then pushes the encoder toward uniform representations, destroying the classification signal. This explains the XLM-R high collapse (0.777) and MuRIL medium collapse (0.786).

Failure Mode 2: Contradictory Gradients. The adversarial head ($\beta = 0.3$, with GRL) and auxiliary head ($\gamma = 0.1$, without GRL) operate on the same $\mathbf{z}^n$ with opposite objectives. With $\beta > \gamma$, the adversarial gradient dominates and the auxiliary head becomes noise.

These findings led us to remove the GRL entirely in v2, keeping only the auxiliary predictor with auto-disable for single-noise-level conditions.

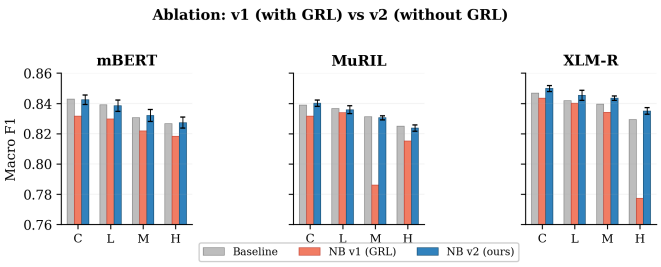

Fig. 4. Ablation across three encoders. Red bars show v1 (GRL) catastrophic failures: XLM-R high (0.777, −5.2 pts from baseline) and MuRIL medium (0.786, −4.5 pts). Blue bars show v2 recovery.

C. Phonetic Weight Analysis

Fig. 5 shows the distribution of phonetic dissimilarity weights ϕ ((7)) across noise levels. The monotonic increase in mean ϕ (low: 0.025, medium: 0.039, high: 0.056) confirms that the weighting correctly assigns stronger contrastive pressure to more distorted examples.

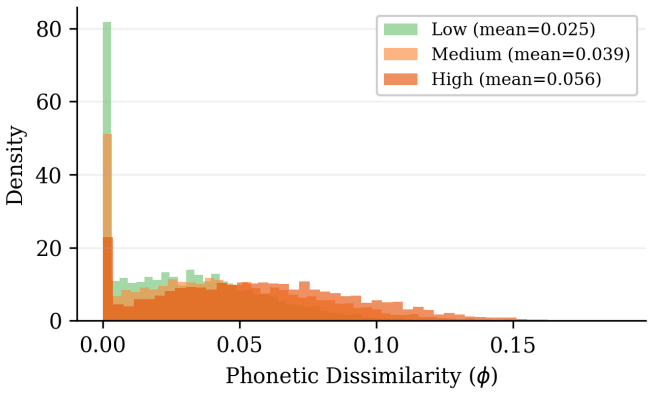

Fig. 5. Distribution of PWNIC weights ϕ by noise level. Higher noise produces higher ϕ, triggering stronger contrastive alignment.

D. Error Analysis

We inspect 50 examples where XLM-R + NoiseBridge differs from the baseline on the high-noise test set. Three patterns emerge: (1) *Noisy Hinglish slurs* (24/50): partially corrupted slurs (e.g., "madarchod" → "madrachd") defeat the baseline but are correctly classified by NoiseBridge, suggesting PWNIC maps corrupted representations near their clean counterparts. (2) *Word boundary errors* (14/50): ASR-style word splits ("chutiya" → "chu ti ya") break tokens into individually non-hateful subwords; the gate appears to down-weight fragments. (3) *English text unaffected* (12/50): minor noise on English text produces near-boundary differences.

VIII. Discussion

When Does NoiseBridge Help? The selective benefit of PWNIC—strong gains on XLM-R but near-zero on mBERT and MuRIL—points squarely at tokenization. XLM-R's SentencePiece operates directly on raw Unicode without case folding or accent stripping, so character-level noise leaves traces that the Jaro-Winkler weighting can exploit. WordPiece models apply normalisation steps that discard exactly those traces before the encoder ever sees them, leaving PWNIC with no phonetic signal to weight. The practical implication is that noise-robustness objectives of this kind need to be matched to an encoder whose tokenizer does not preemptively erase the information being leveraged.

Against Adversarial Disentanglement. After testing gradient reversal with $\beta \in \{0.1, 0.3, 0.5\}$ and $\lambda_{\max} \in \{0.5, 1.0\}$, we found no setting that recovered the baseline, let alone improved on it. The root problem is not a hyperparameter choice: when a training batch contains only one noise level, every adversarial target is the same constant, the discriminator trivially memorises it, and the reversed gradient pushes the encoder toward uniform representations. DANN [2] avoids this because its source and target domains are always distinct—a structural guarantee that does not hold here. Practitioners considering adversarial noise disentanglement in similar settings should account for this degenerate case.

Interaction with Representation Granularity. A broader pattern emerges from these results: robustness to surface-level noise depends on how much sub-character information survives tokenization. WordPiece collapses this information early; byte-level models such as ByT5 [19] preserve it by design, which likely contributes to their reported robustness. The one-epoch ByT5 result in our preliminary experiments (F1 = 0.831) is consistent with this view, though the training cost prevented a full evaluation.

Limitations. Three limitations deserve mention. First, our noise simulation is synthetic: real Hinglish ASR errors include phoneme confusions and language-model biases that uniform random perturbations do not capture. Second, the benchmark skews toward English in some splits, so results may not transfer uniformly to Hindi-dominant speech. Third, the overall headroom is modest—multilingual transformers are already surprisingly robust to character noise—which means even a

well-designed method like PWNIC can only move the needle by a limited amount.

A. Future Work

Several directions follow naturally from this work. The most immediate is replacing synthetic noise with real ASR transcripts: Hinglish speech corpora exist but are not paired with hate-speech labels, so building such a resource—or aligning existing ones—is the clearest path to more ecologically valid evaluation. On the model side, the interaction between tokenisation granularity and PWNIC suggests that byte-level encoders such as ByT5 are worth a full evaluation; our one-epoch results were promising but resource-constrained. Adapting the benchmark and framework to other code-mixed language pairs (Tamil-English, Bengali-English) would test whether the gains generalise beyond Hinglish, where the specific phoneme inventory and romanisation conventions shape noise patterns in ways that may differ substantially. Finally, while we fixed noise probabilities to three discrete levels, learning a noise distribution conditioned on speaker or domain characteristics could make the augmentation more realistic without requiring real ASR data.

IX. Conclusion

We introduced **HindiMix-Noisy**, the first benchmark for evaluating hate speech detection under multi-level ASR-induced noise in Hindi-English code-mixed text, and **NoiseBridge**, a phonetic-aware contrastive framework for improving robustness to such noise. Our results show that while multilingual encoders degrade only modestly under character-level corruption (1–2 F1 points), targeted alignment via PWNIC yields consistent gains for XLM-R (+0.41 average, +0.56 under high noise).

Two findings stand out beyond the numbers. First, robustness to ASR noise is gated by tokenisation granularity: SentencePiece preserves the phonetic residue that PWNIC exploits, whereas WordPiece normalisation erases it before the encoder can act. Second, adversarial gradient reversal—an otherwise successful domain-adaptation tool—breaks down here because batches drawn from a single noise level give the adversarial head a constant target, collapsing the encoder toward uniform representations. Both observations have practical design implications for future noise-robust NLP work.

The dataset, training code, and model checkpoints are publicly released at the link below in hopes that they make it easier for subsequent work to study robustness under realistic code-mixed, speech-driven conditions.[1]

References

[1] W. E. Winkler, "String comparator metrics and enhanced decision rules in the Fellegi-Sunter model," *Proc. Survey Research Methods, ASA*, 1990.

[2] Y. Ganin et al., "Domain-adversarial training of neural networks," *JMLR*, vol. 17, no. 1, pp. 2096–2030, 2016.

[3] T. Davidson, D. Warmsley, M. Macy, and I. Weber, "Automated hate speech detection and the problem of offensive language," in *Proc. ICWSM*, 2017.

[4] B. Mathew et al., "HateXplain: A benchmark dataset for explainable hate speech detection," in *Proc. AAAI*, 2021.

[5] U. Barman, A. Das, J. Wagner, and J. Foster, "Code mixing: A challenge for language identification in the language of social media," in *Proc. First Workshop on Computational Approaches to Code Switching*, 2014.

[6] P. Shah, H. S. Chadha, A. Gupta, and A. Bharadwaj, "Learning to recognize speech in Hindi-English code-mixed environments," *arXiv:2011.12610*, 2020.

[7] M. Mozafari, R. Farahbakhsh, and N. Crespi, "A BERT-based transfer learning approach for hate speech detection," in *Proc. COMPLEX NETWORKS*, 2020.

[8] T. Caselli, V. Basile, J. Mitrović, and M. Granitzer, "HateBERT: Retraining BERT for abusive language detection in English," in *Proc. WOAH*, 2020.

[9] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL*, 2019.

[10] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in *Proc. ACL*, 2020.

[11] S. Khanuja et al., "MuRIL: Multilingual representations for Indian languages," *arXiv:2103.10730*, 2021.

[12] A. Pratapa et al., "Language modeling for code-mixing," in *Proc. ACL*, 2018.

[13] S. Khanuja et al., "GLUECoS: An evaluation benchmark for code-switched NLP," in *Proc. ACL*, 2020.

[14] A. Bohra, D. Vijay, V. Singh, S. S. Akhtar, and M. Shrivastava, "A dataset of Hindi-English code-mixed social media text for hate speech detection," in *Proc. PEOPLES*, 2018.

[15] T. Mandl et al., "Overview of the HASOC subtrack at FIRE 2021," in *Proc. FIRE*, 2021.

[16] A. Piktus et al., "Misspelling oblivious word embeddings," in *Proc. NAACL*, 2019.

[17] Y. Belinkov and Y. Bisk, "Synthetic and natural noise both break neural machine translation," in *Proc. ICLR*, 2018.

[18] S. Eger et al., "Text processing like humans do: Visually attacking and shielding NLP systems," in *Proc. NAACL*, 2019.

[19] L. Xue et al., "ByT5: Towards a token-free future with pre-trained byte-to-byte models," *TACL*, vol. 10, pp. 291–306, 2022.

[20] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple contrastive learning of sentence embeddings," in *Proc. EMNLP*, 2021.

[21] P. Khosla et al., "Supervised contrastive learning," in *Proc. NeurIPS*, 2020.

[22] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv:1807.03748*, 2018.

[23] Y.-H. Chang and Y.-N. Chen, "Contrastive learning for improving ASR robustness in spoken language understanding," in *Proc. Interspeech*, 2022, pp. 3458–3462.

[24] M. Zampieri et al., "Predicting the type and target of offensive posts in social media," in *Proc. NAACL*, 2019.

[25] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. ICML*, 2020.

[26] X. Zhang, J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification," in *Proc. NeurIPS*, 2015.

[27] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[1] https://github.com/Abhijit26918/hindiMix-noisy